Doge-SAGE

Stateful Agent Governance and Execution for Scientific Communities

Doge project / V5

1 A bot that stays

Scientific questions, shared materials and casual exchanges coexist in the same group.

A persistent scientific companion must coordinate people, tools and history as their tasks overlap.

Design objective
One companion; many overlapping tasks.

Three architectural contributions

A A shared capability substrate
Retrieve only relevant operations. Natural language and explicit commands converge on shared handlers and group-level availability.

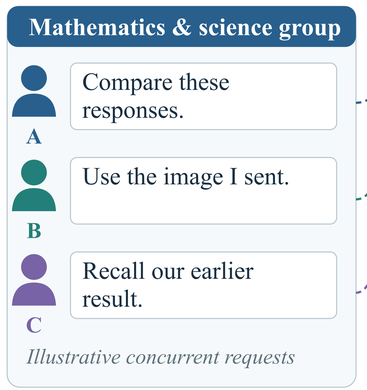
B State with distinct lifetimes
Keep identity stable while selecting social cues, working context, archival evidence and sender-scoped materials for each turn.

C Concurrent work, shared continuity
Give each event its own runner; merge its new history into the latest shared state at a short commit boundary.

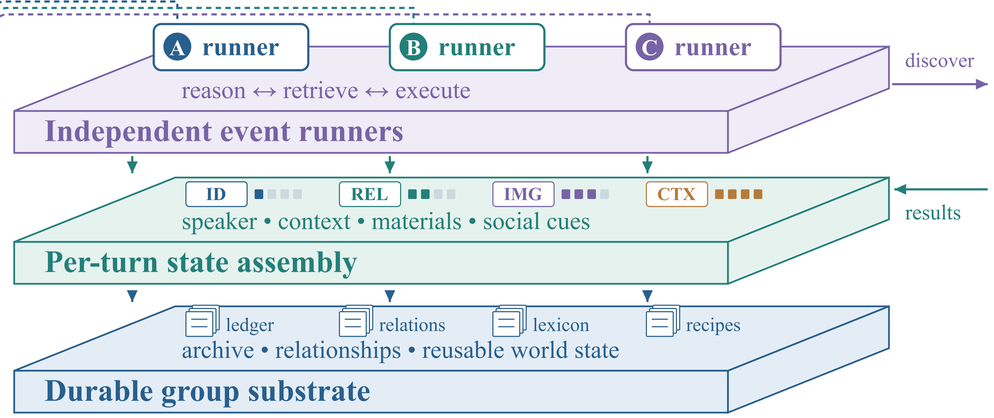
Capability-level output contracts
Facts · **Compute** · **Simulate** · **Create**

2 Inside the long-lived group companion

Shared group space



Event-local execution above persistent state



A heterogeneous workbench

- **Math & computation** — Symbolic math · Fourier · control systems
- **Science & evidence** — Literature · biology · crystals · astronomy
- **Language & creation** — Tangut · phonology · RRPL · Cthuvian
- **Play & social worlds** — Persistent games · alchemy · social media

Select assets; compose the response
Explain → Figure → Interpret

Event admission → scoped state + capability access → LLM / tools → selective rich output • archive and state persist across turns

Pre-inference admission control request → admit → model + tools; explicit exhaustion intent → local rejection. Rejected branch: zero model tokens

A Retrieve a leaf; retain one execution path

Capability registry

Relevant + available
canonical ID · arguments
kind · source · fallback

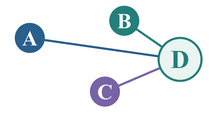
36 families · 281 leaves

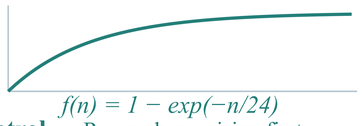
Natural-language discovery — **Explicit command** → **Shared domain handler**

One registry grounds discovery and usage; group module state constrains both invocation paths.
Two entry paths; one shared capability contract.

B Assemble context without erasing identity

Stable identity + policy + capability decoder
Working history + current message
Social cues + material / capability hints





Research: precision first

Factorized persona control
relationship
transient affect
→ g → **warmth** · **playfulness** · **restraint**

C Parallel inference, short history commits

Merge new tails under lock; inference stays parallel.

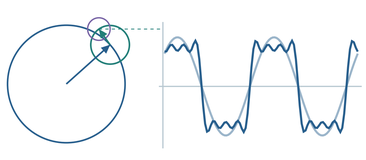
$Hnext = Hlatest + \Delta event$

Local admission Load abuse: execution / repetition / load patterns → reject — Benchmarks: rule miss → shape gate → int8 score
Pre-provider rejection Hashed character n-grams · 2 KiB int8 weights · real-work context bypasses the statistical scorer

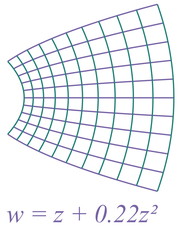
3 A visual scientific workbench

Harmonic synthesis, conformal maps & control

$y(t) = \Sigma\ sin(kt) / k,\ k = 1, 3, \ldots$

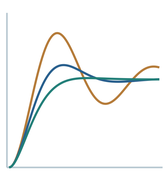


Harmonics → waveform



$w = z + 0.22z^2$

$G(s) = 1/(s^2+2\zeta s+1)$



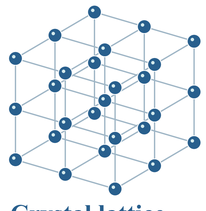
Damping sweep
$\zeta = 0.2, 0.5, 0.8$

Symbolic mathematics, Fourier epicycles and control models turn abstract mechanisms into inspectable visual experiments.

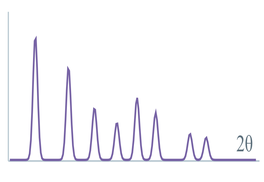
TeX · Typst · TikZ · Mermaid · SVG / PDF

4 From structure to scientific evidence

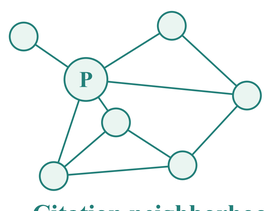
Structures, diffraction & connected literature



Crystal lattice



Model diffraction



Citation neighborhood
DOI · metadata · full text

CIF / XRD, molecular and protein resources, literature graphs and astronomy connect physical models with source-backed evidence.

Chaoli: new topics / replies → proactive notifications
Successful delivery → advance notification state

5 Language & persistent creative worlds

Cthuvian: a lexicon that grows through use

known · missing · known · missing → **Proposal batch**

Reuse entries — **Validate → commit together**

Tangut	query	lexicon	translation
Phonology	character	lookup	history / dialect
RRPL	syntax	explain	render

Concept alchemy



Saved products become reusable ingredients.

Persistent symbolic state: validated lexicons + reusable recipes

Major generations V2 Rules → V3 Domains → V4 Plugins → **V5 Group agent runtime**

Capability space 36 families · 281 canonical leaves · 576 invocation forms

Project: github.com/ChouYuanjue/doge-repo • Architecture built on AstrBot / OneBot primitives.

September 2026 dossier. Scientific visuals are illustrative; curves use declared mathematical models.